School of Geography and Environment, Jiangxi Normal University

Monitoring Pine Wilt Disease-Infected Trees Using UAV Remote Sensing and Deep Learning Algorithms

A Case Study of the Mountainous Region of Southern Jiangxi

Competition team: D626

Advisers: Xu Jian, Huang Shiming

Team members: Sun Zhenghe, Zheng Leran, Wu Shuyi, Zheng Yajie

Contents

01 Project Overview

02 Design Approach

03 Implementation

04 Results and Analysis

01 Project Overview

1.1 Background

A devastating forest disease

Pine wilt disease is one of the world's most dangerous and destructive forest diseases. It thrives in many environments, affects numerous host species, spreads rapidly, causes high mortality in infected pines, and is difficult to control. Infected pines can die within 40 days. Without treatment, entire pine forests can be destroyed within 3–5 years, seriously threatening China's ecological security. Research on technologies for preventing and controlling pine wilt disease is therefore essential.



Figure: Infected trees

Wider use of UAV remote sensing

Traditional monitoring relies solely on manual ground surveys, which are time-consuming, labor-intensive, and inefficient. As UAV remote sensing becomes more widespread, its use in pest and disease monitoring has become a research focus. Combining it with deep learning is an inevitable direction for forest pest and disease control. This project proposes a method using UAV remote sensing and deep learning to identify pine wilt disease-infected trees rapidly and automatically.

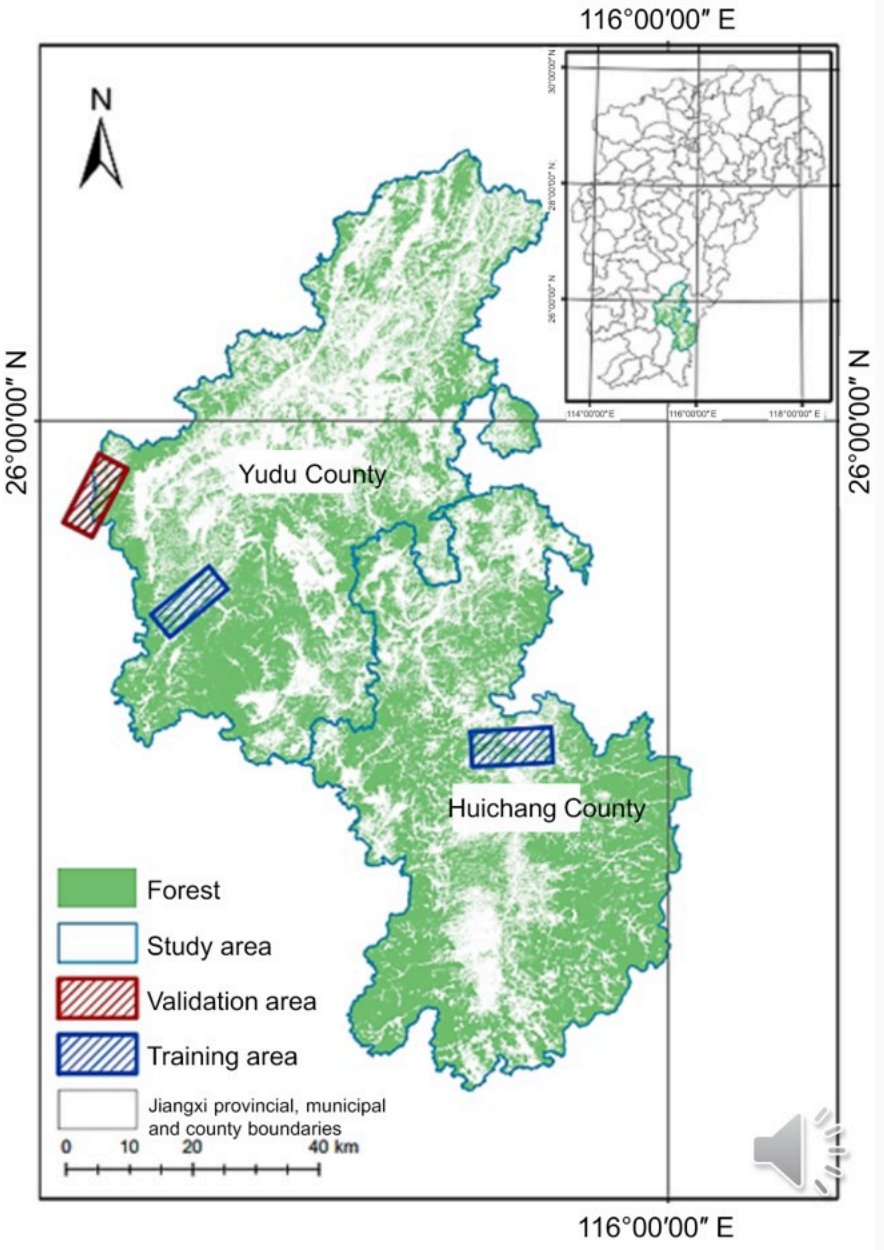
1.2 Study Area

Location: Ganzhou, southern Jiangxi Province, 113°54′–116°38′ E and 24°29′–27°09′ N.

Terrain: At the junction of the Nanling, Wuyi, and Zhuguang mountain ranges. The area lies in the hills south of the Yangtze River and has typical hilly and mountainous terrain.

Climate: On the southern edge of the mid-subtropical zone, with a subtropical monsoon climate, abundant rainfall and sunshine, a large daily temperature range, and a long frost-free season.

Pine forest resources: Ganzhou has extensive pine forests covering about 15 million mu, approximately one-third of the province's total.



1.3 Survey of Infected Pines

Pines are a genus of evergreen trees in the family Pinaceae. They are widespread across the Northern Hemisphere and are major sources of timber and resin worldwide. In China, they occur across almost the entire country.

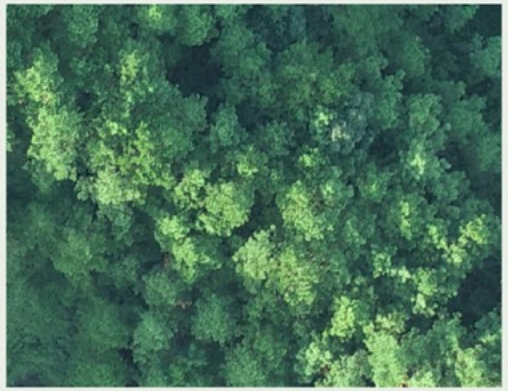
Masson pine is native to Jiangxi. Its rapid growth, stress tolerance, and adaptability make it an important pioneer species for timber plantations and ecological afforestation in the province. Masson pine accounts for 78.36% of Jiangxi's pine forest area. Its extensive, widespread, contiguous stands provide the host conditions that allow pine wilt disease to spread through the mountainous region of southern Jiangxi.



Figure: Pine trees

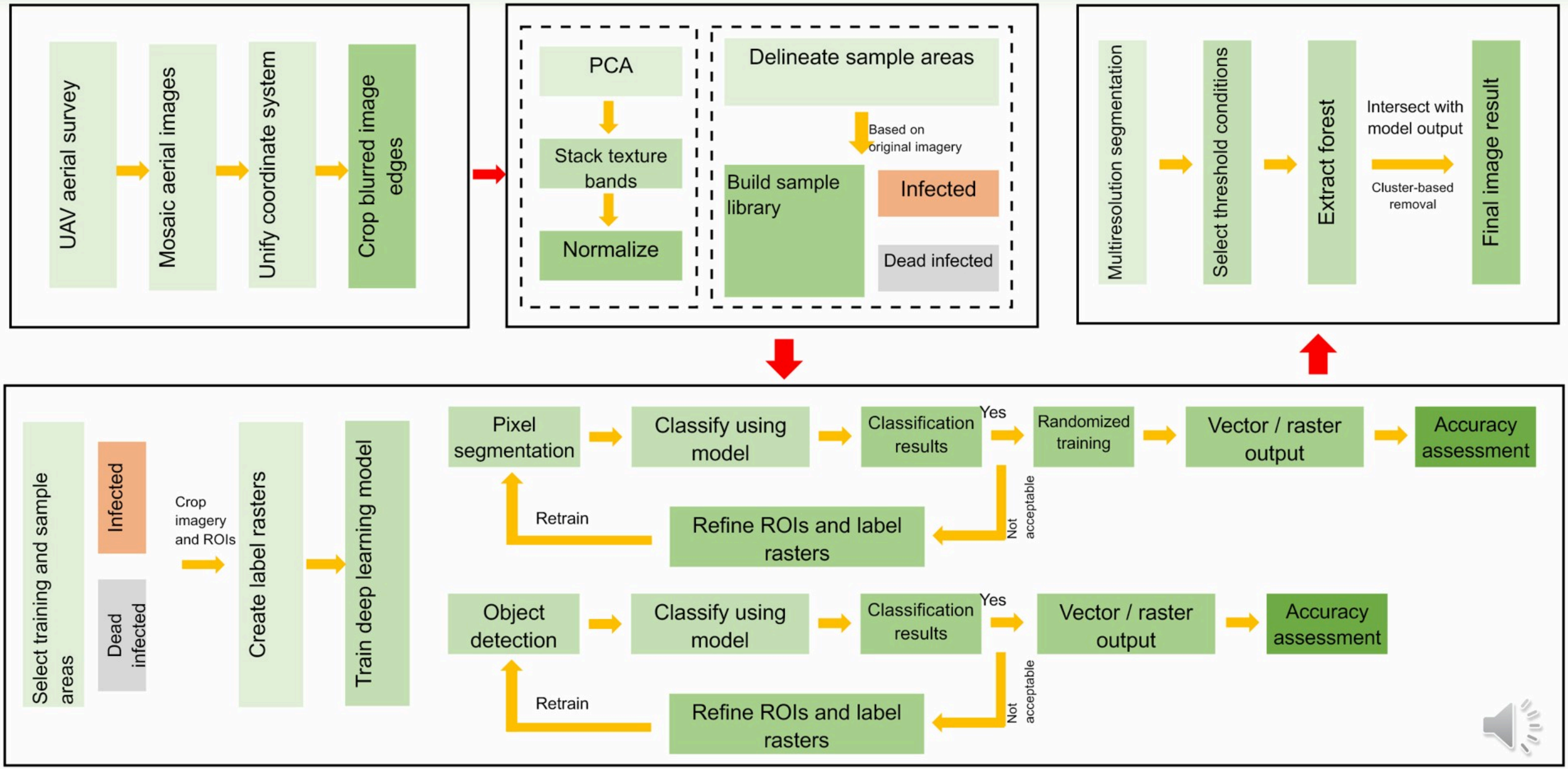
1.4 Survey of Infected Pines

Symptoms at Different Stages of Infection

Healthy pine	Early infection	Middle stage	Late stage
Normal green needles and vigorous branches	Needles gradually become paler and yellow	Branches and needles dry out and turn reddish brown	The tree dies completely and sheds all needles. Its exposed, pale branches form a firework-like crown.
			

02 Design Approach

2.1 Technical Workflow



03 Implementation

3.1 Training and Validation Area Selection

Huichang County:

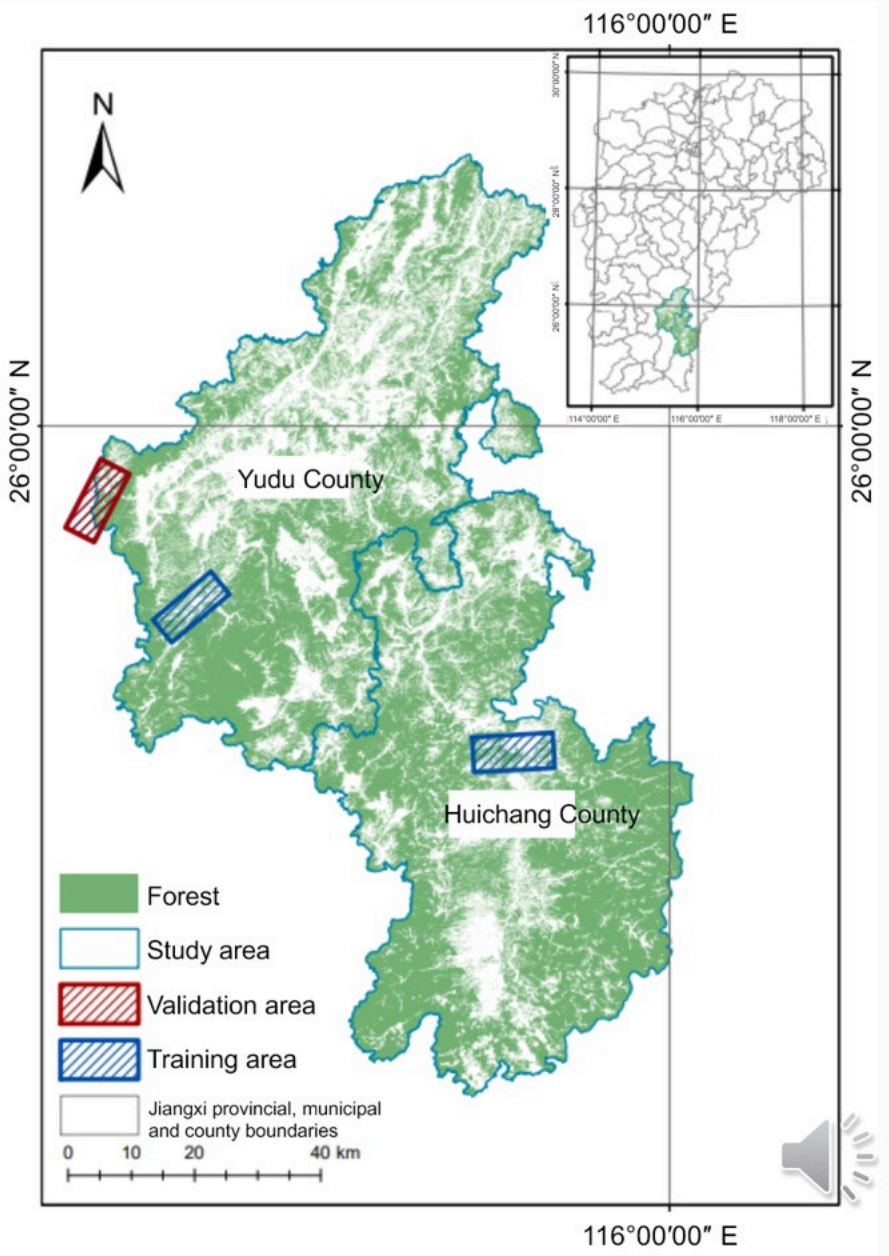
The outbreak is severe and still developing, making the area suitable for long-term monitoring. Comprehensive imagery is also available, providing sufficient training samples.

Pine wilt disease was first detected in Zhanggong District, Huichang County, in 2003 and has now entered a full outbreak phase. The 2020 survey recorded 20,325 mu of affected forest across six townships in the county (Huichang Government public information).

Yudu County:

Pine wilt disease affects a large and growing area.

The Japanese pine sawyer beetle, a vector of pine wilt disease, is widespread and affects a large area, with worsening damage in some locations. Local outbreaks of the Masson pine caterpillar also cause severe damage (Yudu Government public information).



3.2 Data Overview

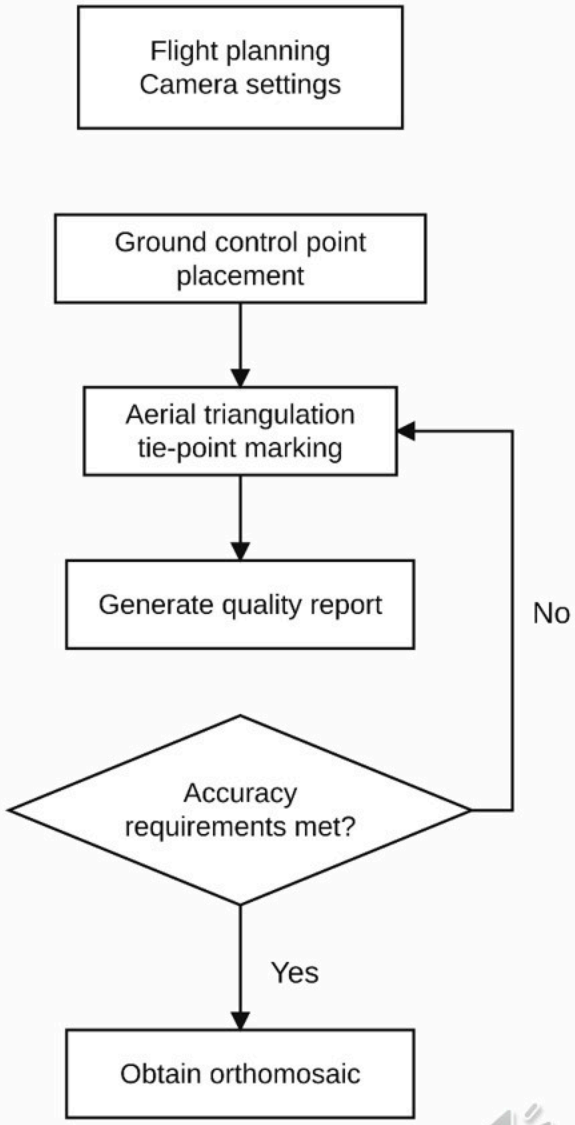
UAV Image Acquisition

1. Select the UAV survey area. Choose representative areas and divide them into training, validation, and test sample areas, plus field verification areas for assessing model accuracy.

2. Plan UAV flight routes. Account for the distribution and elevation of Masson pine stands, UAV flight range and altitude, forward overlap, and side overlap when planning routes.

3. Conduct UAV aerial photography. Following GB/T 23478-2009, Technical Regulations for General Survey and Monitoring of Pine Wilt Disease, survey twice a year during the spring and autumn survey periods (March–June and August–October). Fly at 9:00 a.m. in clear weather to avoid clouds, thunderstorms, and other weather interference. In autumn, avoid the period when broadleaf foliage changes color to reduce its interference with detection.

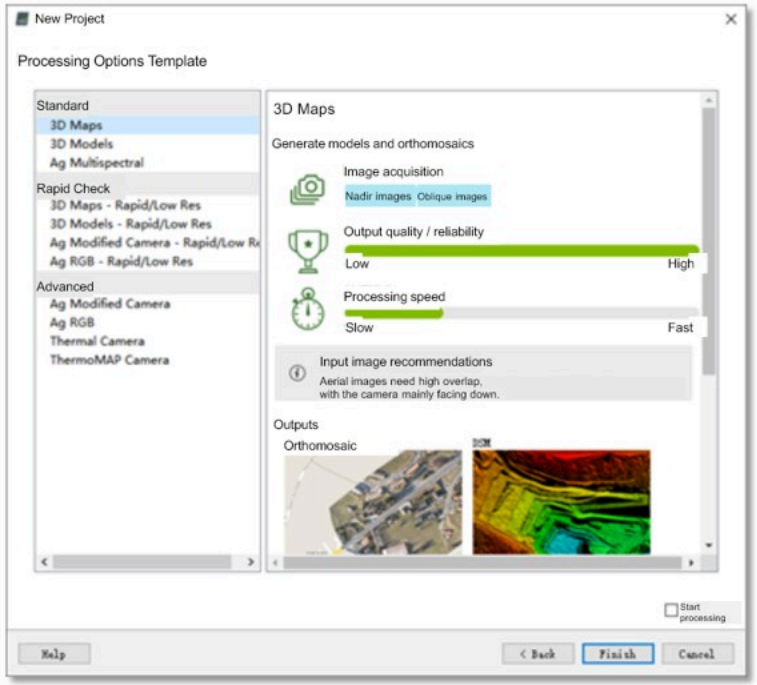
4. UAV specifications. DJI Phantom 4, flying at an altitude of 150 m.

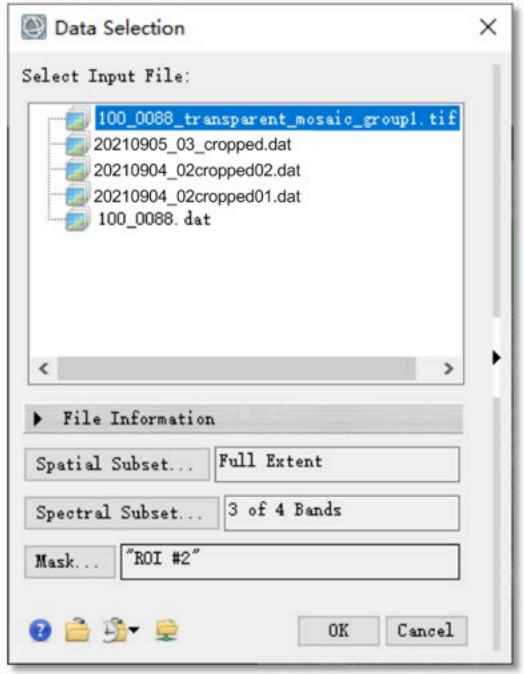


UAV image processing workflow

3.3 Data Preprocessing

Aerial Image Mosaicking





Aerial images were mosaicked using Pix4Dmapper.

The imagery does not require ground control points. Import the aerial images and select the standard 3D Maps template to create the mosaic. When processing is complete, save and export the imagery from the Mosaic Editor.

Import the mosaic into ENVI and draw an ROI based on edge clarity and distortion. Use the ROI as a mask and export the data in ENVI format. Considering image color and the four bands in the original imagery, export only the first three bands.

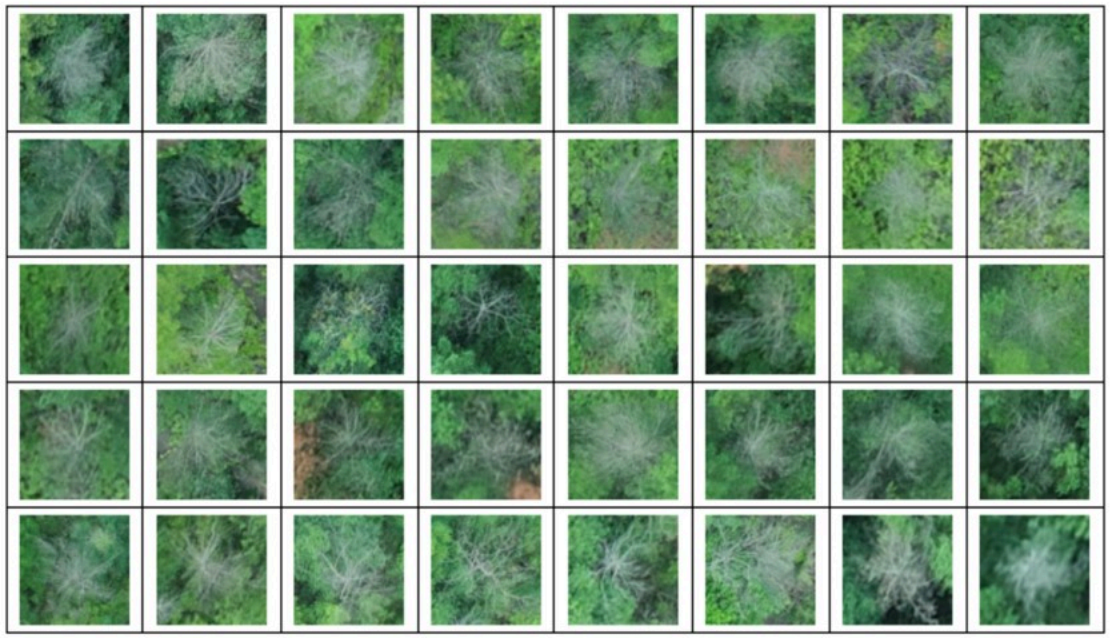
3.4 Sample Selection Criteria

Type	Visual Interpretation Criteria	Notes	Example
Infected trees	Color: yellowish brown, reddish brown, or brown Shape: firework-like crown Texture: compact	Needles begin to fall in the middle and late stages	
Dead infected trees	Color: gray Shape: firework-like crown Texture: open, with clearly visible branches	All needles have fallen. Delineate only the central portion, keeping the boundary as small as possible.	

3.4 Sample Dataset Creation



Examples of infected-tree samples



Examples of dead infected-tree samples

During sample selection, we included as many samples as possible from the training areas and carefully traced their boundaries so the deep learning model could learn their spectral and shape features.

3.4 Sample Examples (Infected Trees)

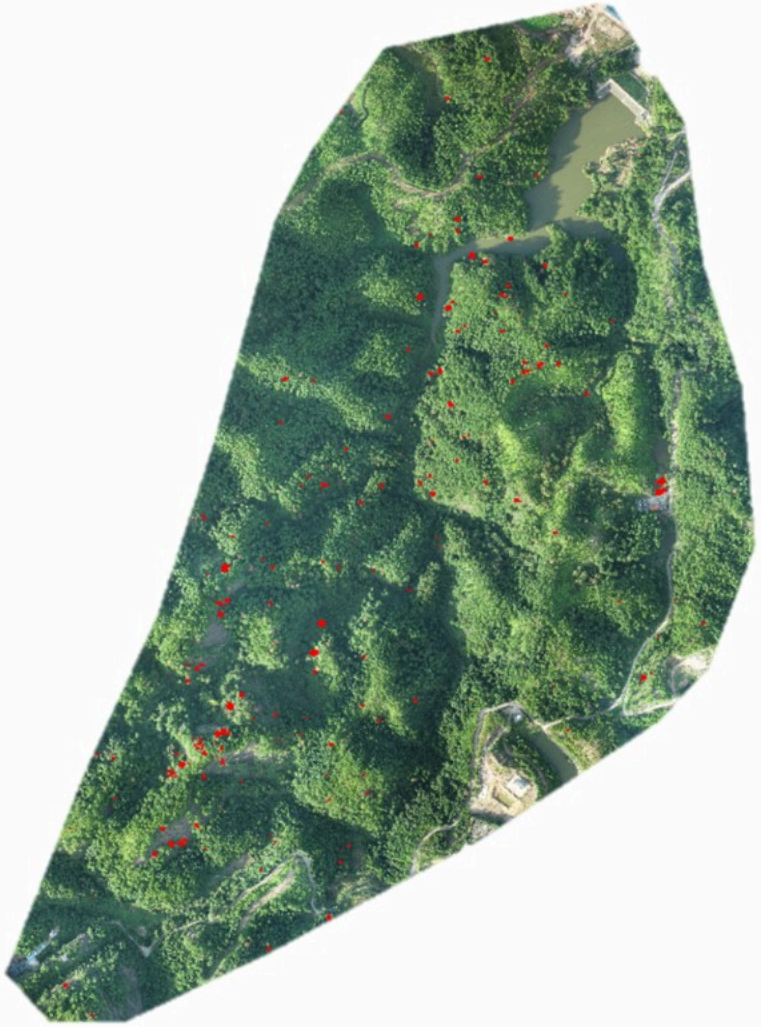


Image 1: 183 samples



Image 2: 45 samples



Image 3: 49 samples

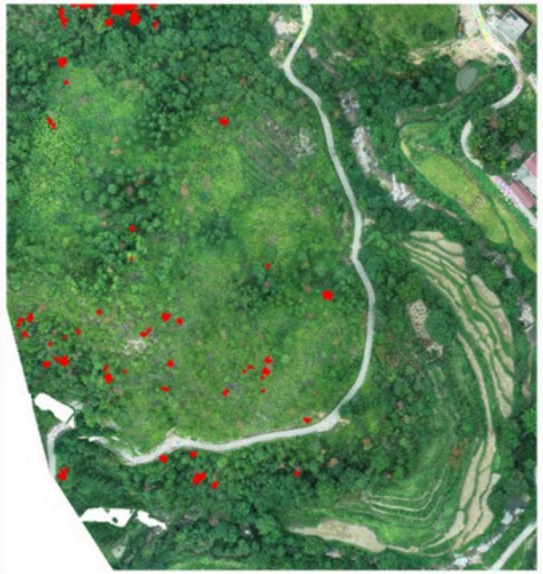


Image 4: 53 samples

Total infected-tree and dead infected-tree samples: 1,143

3.4 Sample Overview

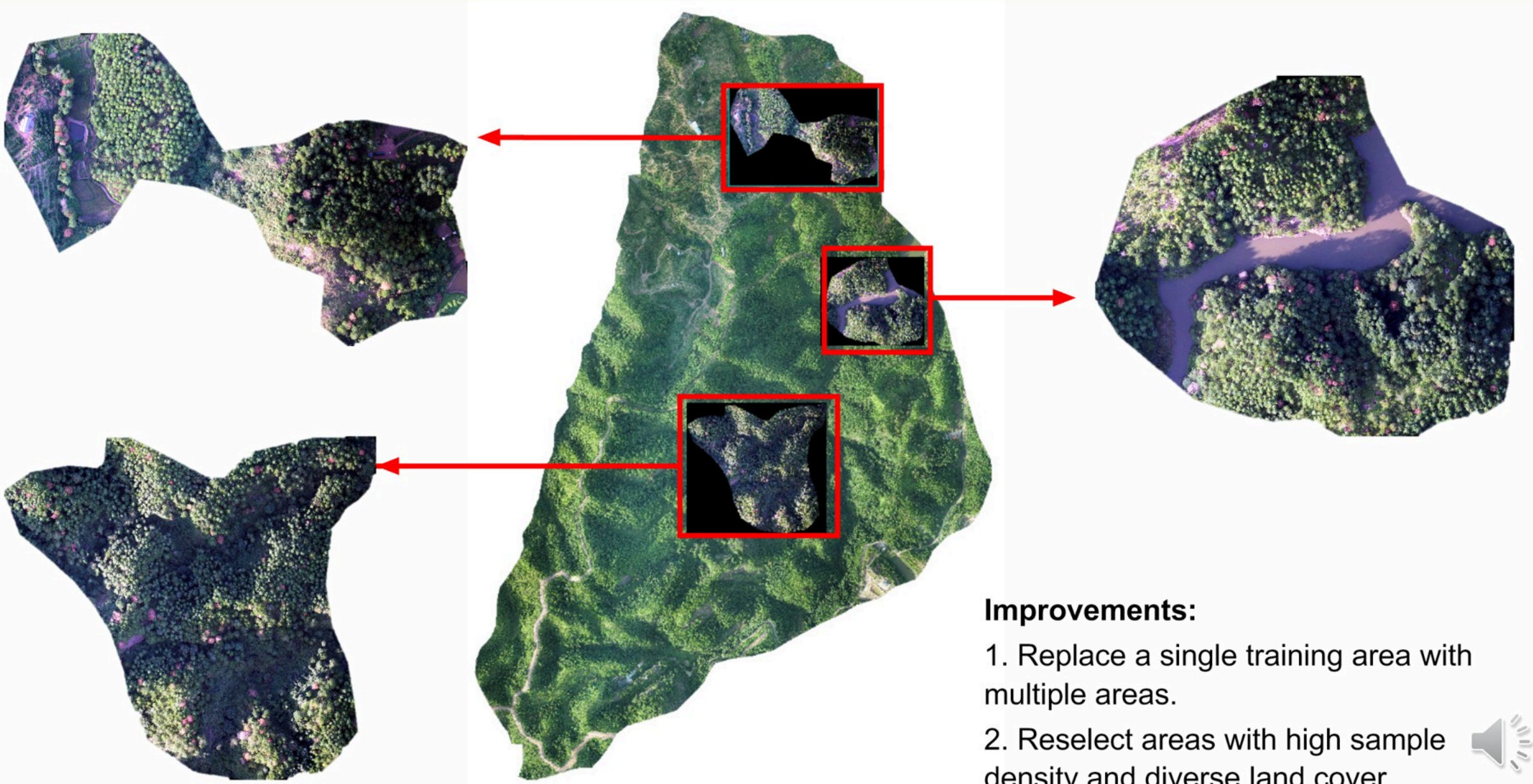
Area	Infected-tree samples
shigongyan_01	35
shigongyan_02	28
shigongyan_03	52
202109503	97

Area	Dead infected-tree samples
shigongyan_01	14
shigongyan_02	12
20210904202	23

Using consistent selection criteria, we delineated infected-tree and dead infected-tree samples in each area with ENVI's ROI drawing tool to build the sample database.

We then cropped the training areas and corresponding samples according to the principle of high sample density and diverse land cover. Sample counts are shown

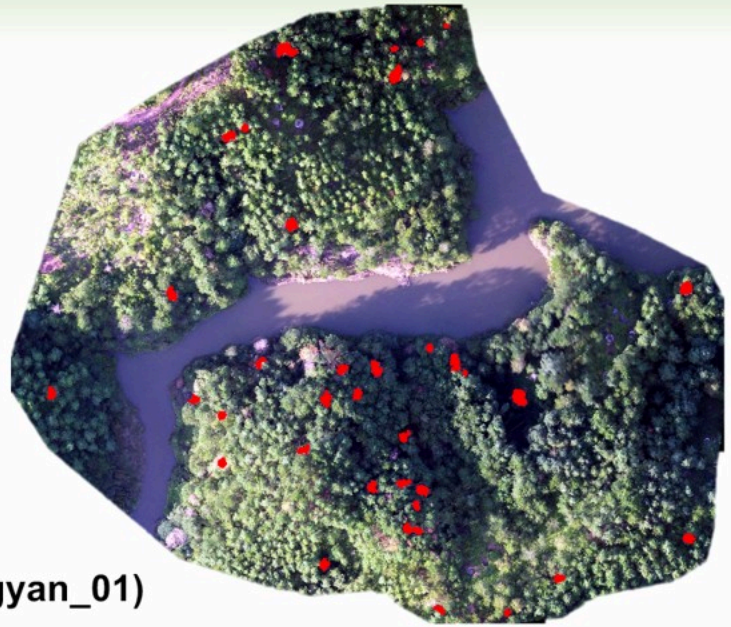
3.1 Training and Validation Area Selection



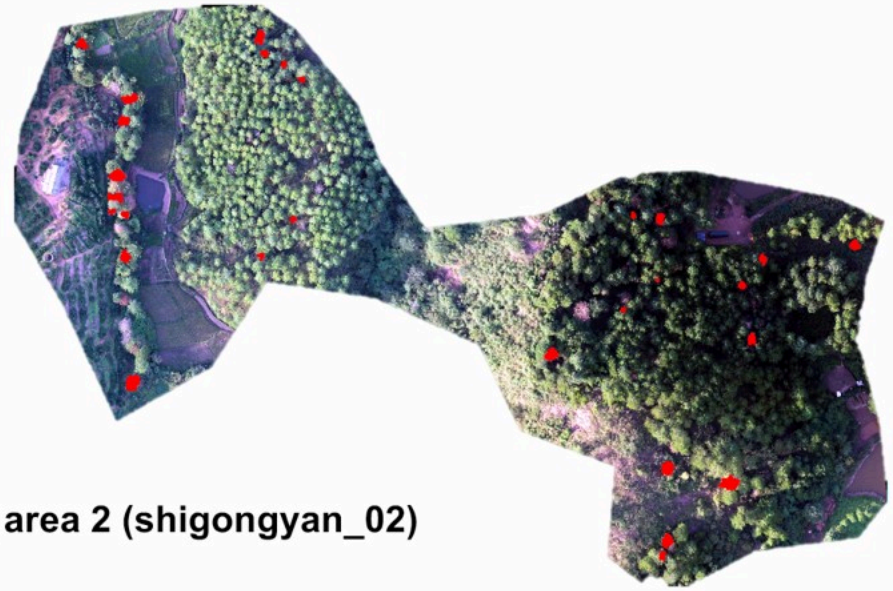
Improvements:

1. Replace a single training area with multiple areas.
2. Reselect areas with high sample density and diverse land cover.

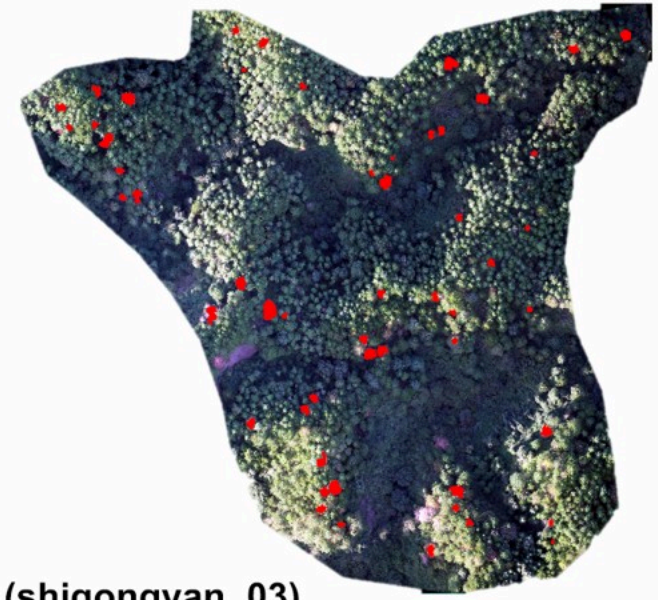
Training Area Examples (Infected Trees)



Training area 1 (shigongyan_01)



Training area 2 (shigongyan_02)



Training area 3 (shigongyan_03)



Training area 4 (202109503)

3.5 Feature Set Preparation

Feature Overview

Visible-light bands from the original imagery and texture features were used to extract infected trees. The eight texture statistics were mean, variance, homogeneity, contrast, dissimilarity, entropy, angular second moment, and correlation.

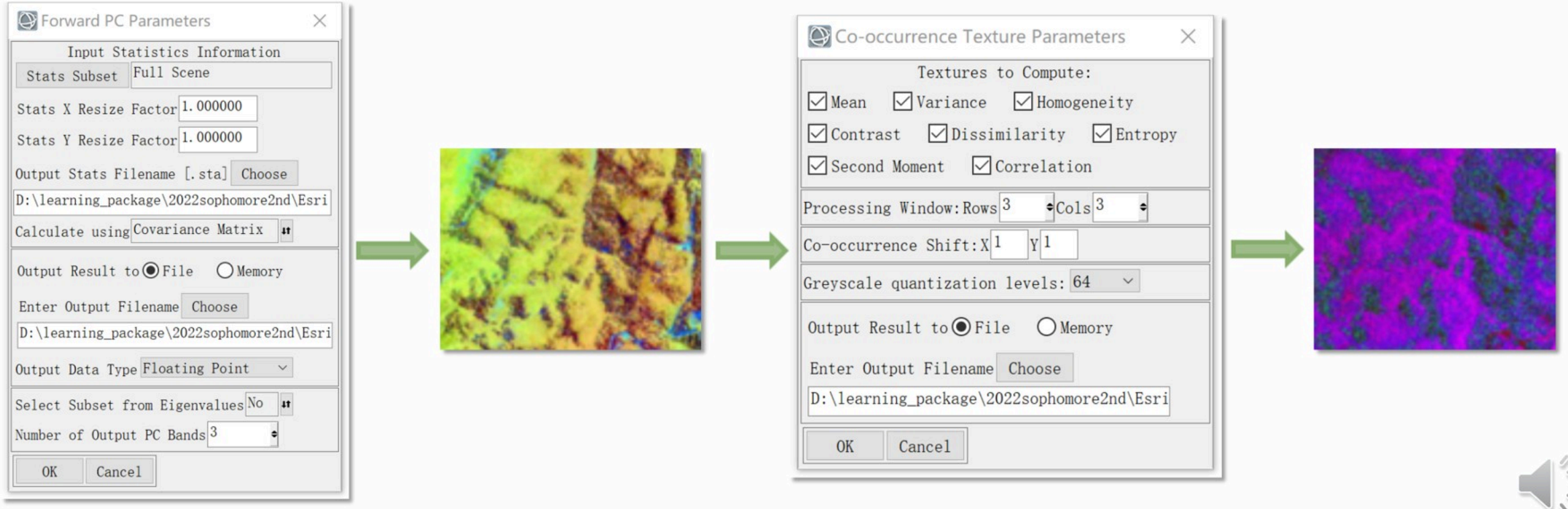
After extracting and stacking all features, we normalized them to remove the effects of differing units and make the indicators comparable.

Feature type	Feature name	Formula
Surface reflectance	Visible-light bands	
Texture features	Mean	$\text{Mean} = \Sigma_{i,j=0}^{N-1} ip(i,j) = u$
	Variance	$\text{Variance} = \Sigma_{i,j=0}^{N-1} p(i,j)\,(i-u)^2$
	Homogeneity	$\text{Homogeneity} = \Sigma_{i,j=0}^{N-1} \frac{P(i,j)}{1+(i,j)^2}$
	Contrast	$\text{Contrast} = \Sigma_{i,j=0}^{N-1} p(i,j)\,(i-j)^2$
	Dissimilarity	$\text{Dissimiliarity} = \text{Contrast} = \Sigma_{i,j=0}^{N-1} p(i,j)\lvert i,j\rvert$
	Entropy	$\text{Entropy} = \Sigma_{i,j=0}^{N-1} p(i,j)(-\ln p(i,j))$
	Second Moment	$\text{Second Moment} = \Sigma_{i,j=0}^{N-1} p\,(i-j)^2$
	Correlation	$\text{Correlation} = \Sigma_{i,j=0}^{N-1} p(i,j)\frac{(i-u_i)(j-u_i)}{\sqrt{\sigma_i^2\sigma_j^2}}$

3.5 Feature Set Preparation

Texture Feature Extraction

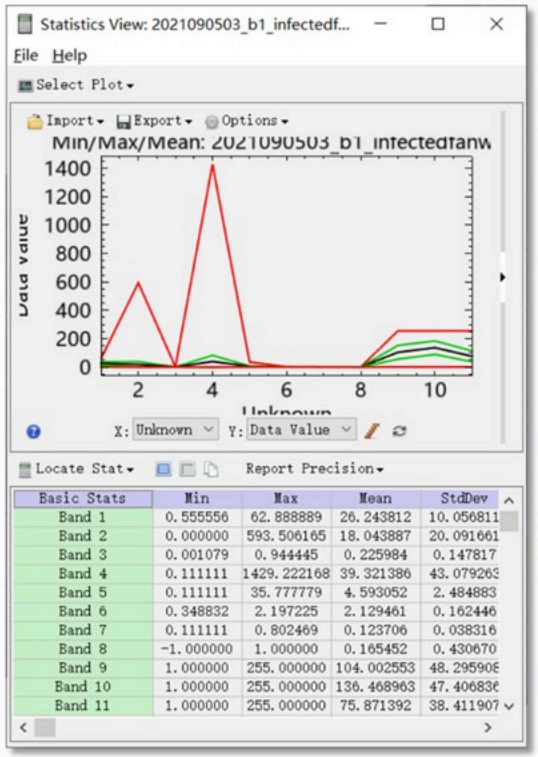
Load the imagery into ENVI, perform principal component analysis, and extract the first principal component. This reduces dimensionality and data volume while retaining as much information as possible. Then use the Co-occurrence Measures statistical filtering tool to extract texture features and stack the resulting images.

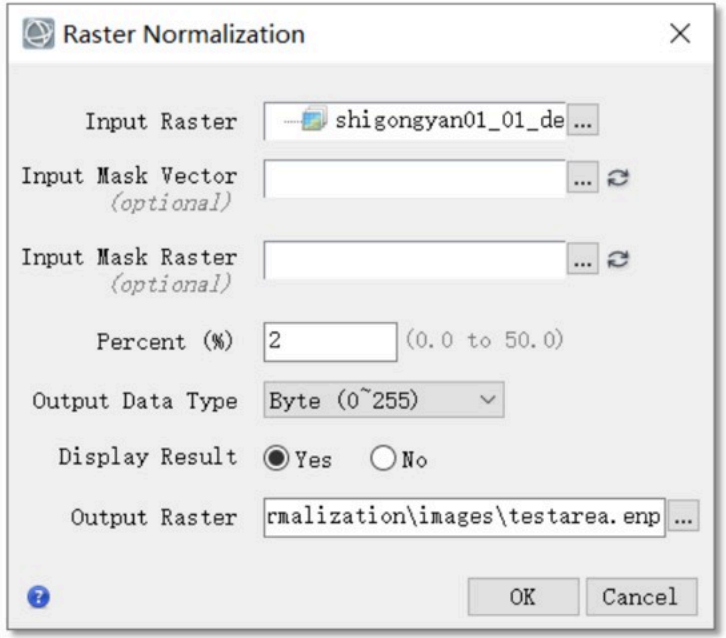


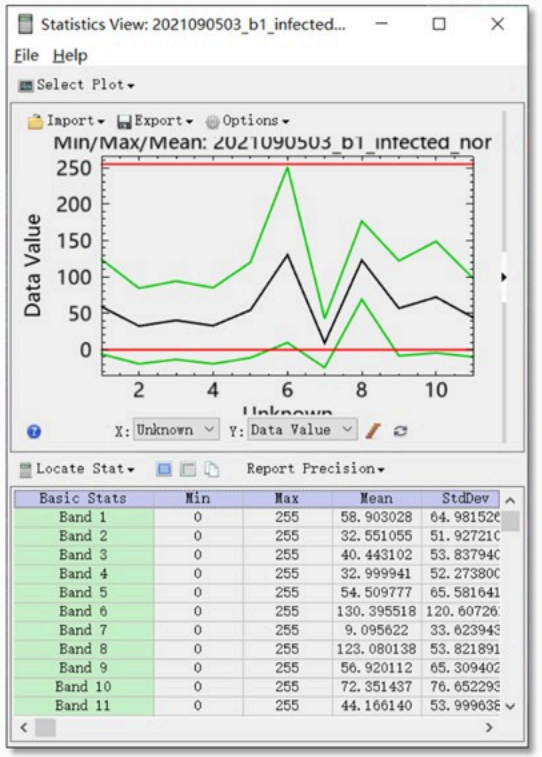
3.5 Feature Set Preparation

Normalization

Use ENVI's Raster Normalization extension to map image values to a specified range and convert dimensional quantities to dimensionless values, improving the model's convergence speed and accuracy.







3.6 Deep Learning

ENVI's Deep Learning module provides two methods:

Pixel Segmentation and Object Detection. Each has advantages and disadvantages, so we built both types of model and compared them.



Pixel segmentation produces irregular patches

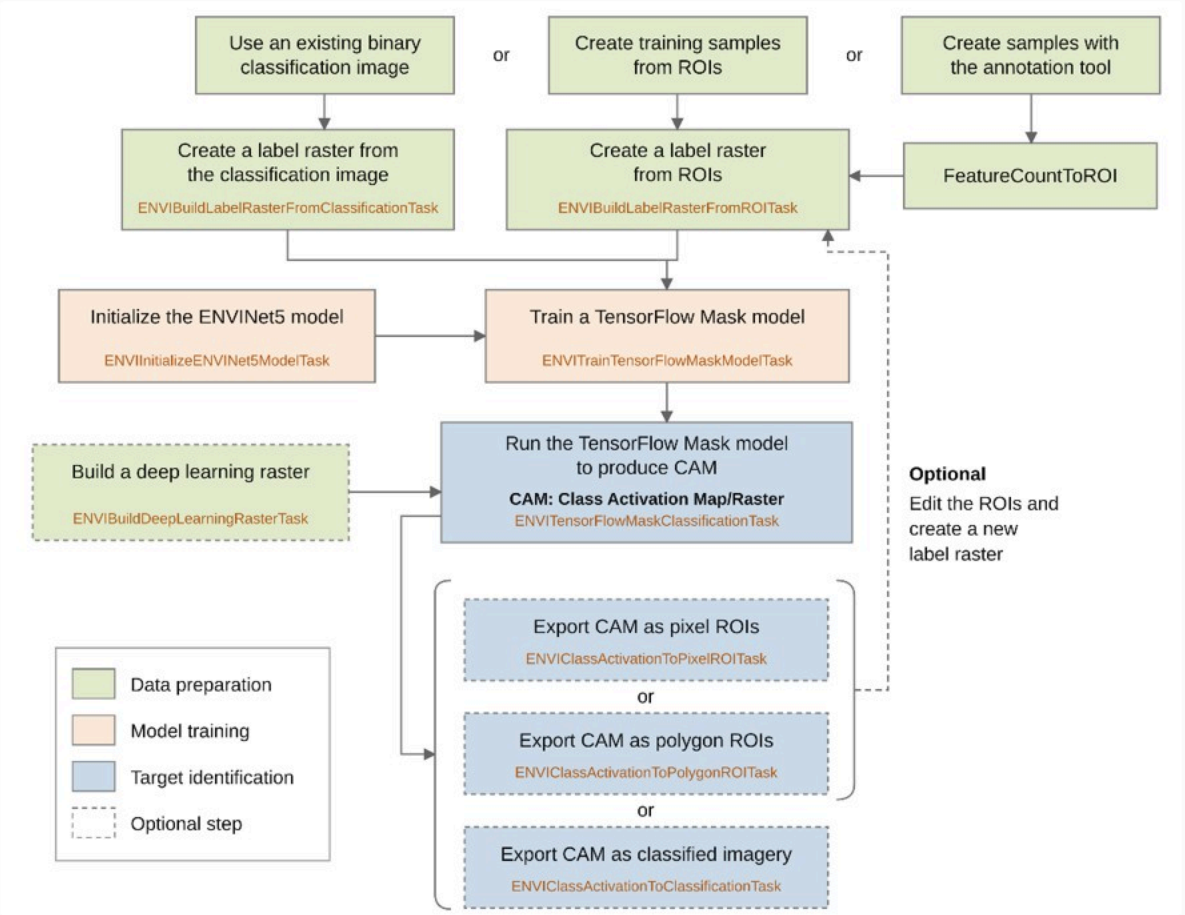


Object detection produces rectangles

3.6.1 Deep Learning with Pixel Segmentation

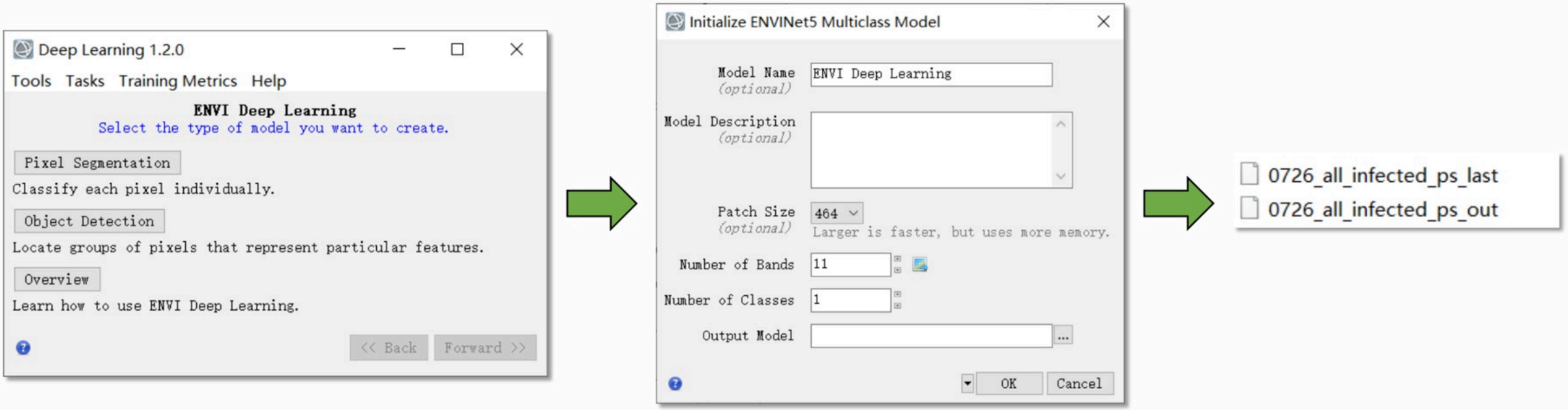
ENVI Deep Learning is a TensorFlow-based tool for remote sensing image classification and object recognition. Its ENVINet5 architecture is based on U-Net, developed by Ronneberger, Fischer, and Brox (2015). Like U-Net, ENVINet5 is a mask-based encoder–decoder architecture that classifies every pixel in an image.

Pixel segmentation identifies pixels belonging to features of interest and assigns class labels to them. In addition to locating objects, it provides information about their shape, area, and other properties.



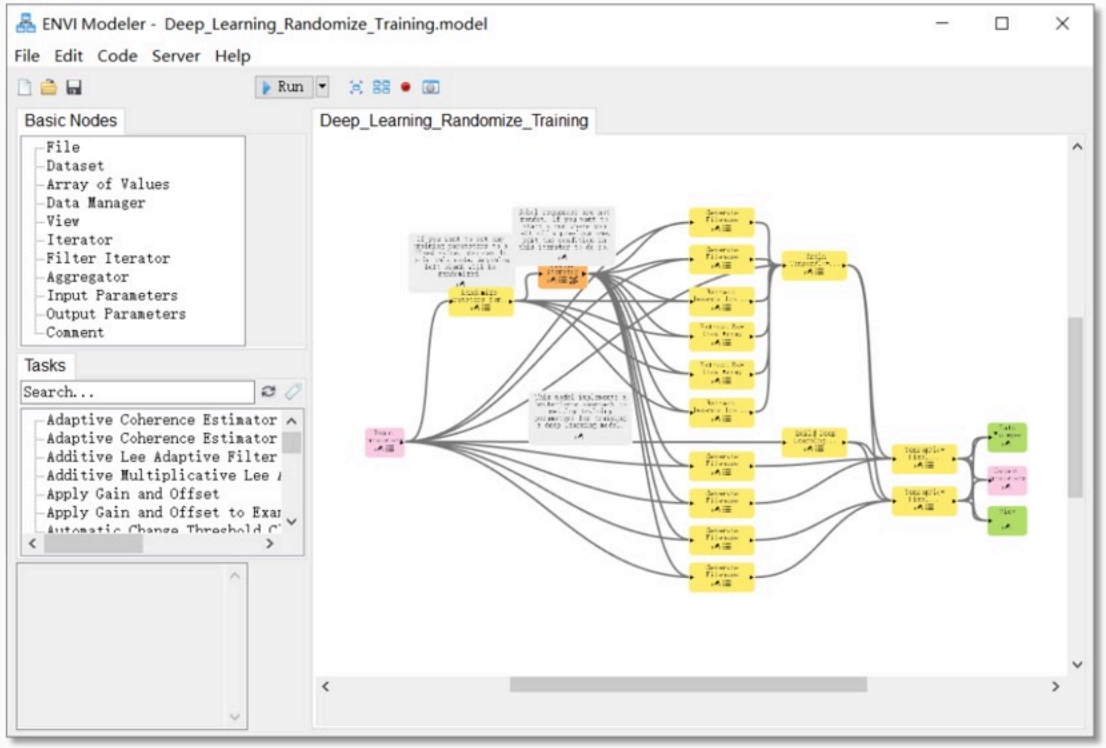
3.6.1 Deep Learning with Pixel Segmentation

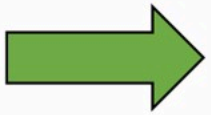
The Deep Learning Guide Map wizard quickly creates and trains models. Initialize a model with Patch Size set to 464, 11 bands, and one class. Keep epochs and other training parameters at their defaults. Use sample ROIs to create input label rasters for three training areas and build two models. Select the model with better accuracy and visual results for randomized parameter-group training.

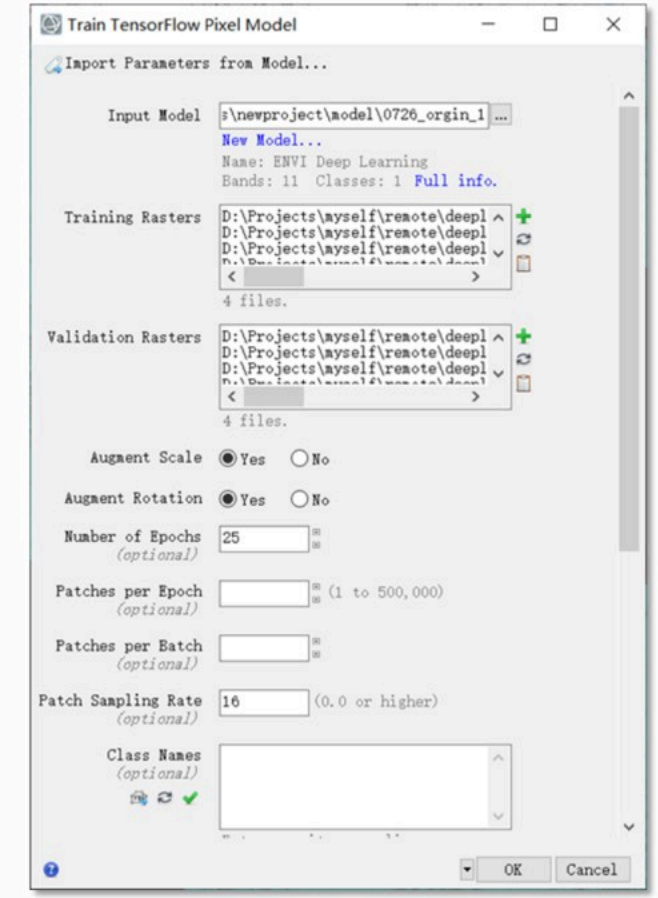


3.6.1 Deep Learning with Pixel Segmentation

Open the randomized parameter-group training module and select Run. Input the best model, label rasters, and imagery for classification to visualize model performance. The module automatically assigns different parameter values for training and iteratively optimizes them to find the best model.

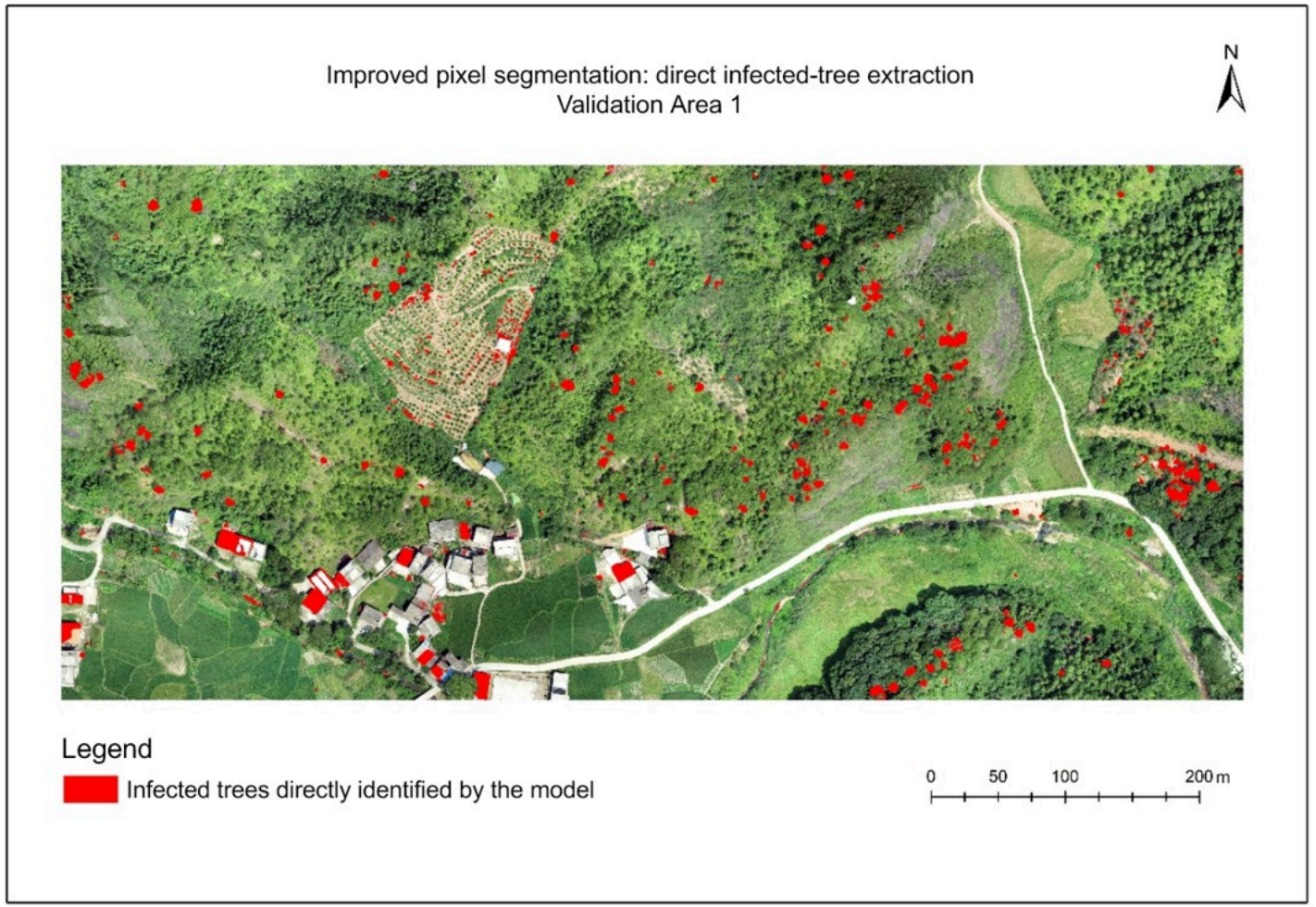






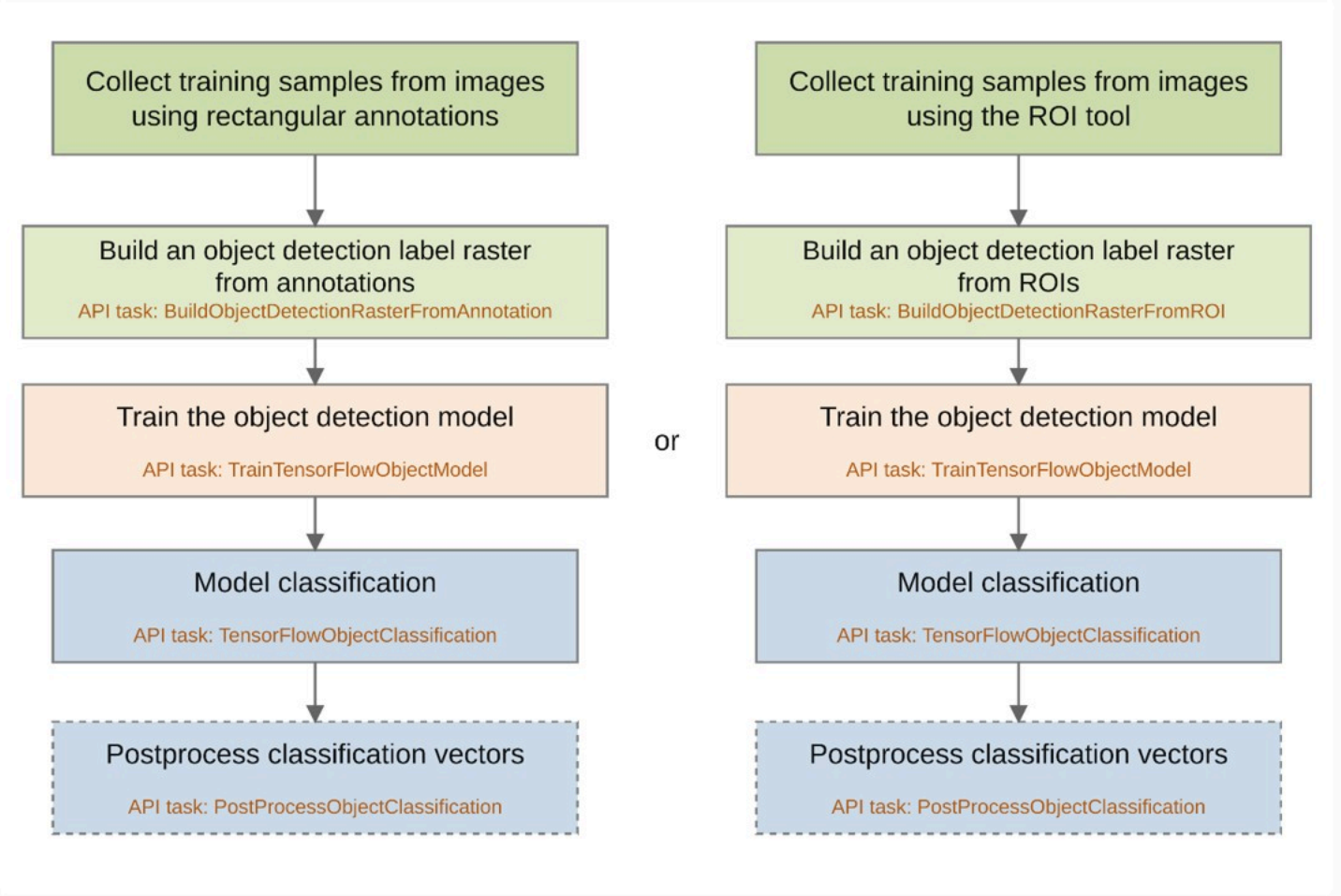
3.6.1 Deep Learning with Pixel Segmentation

Randomized parameter-group training produced 25 models (BEST00–BEST24). Comparing their reported metrics identified BEST12, BEST13, and BEST15 as the strongest. We then extracted trees in the validation area and compared accuracy against the original samples, treated as ground truth. BEST12 gave the best actual extraction results.



3.6.2 Deep Learning with Object Detection

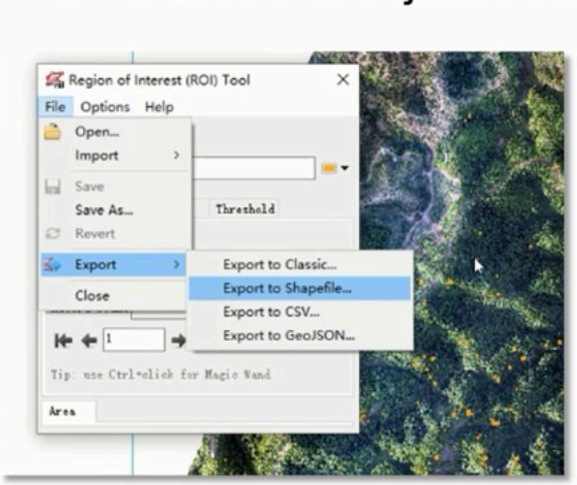
ENVI Deep Learning 1.2 introduced object detection. In theory, it can learn spatial, spectral, and texture features more effectively than pixel segmentation and handle overlapping objects better. In practice, we applied both methods and compared them.

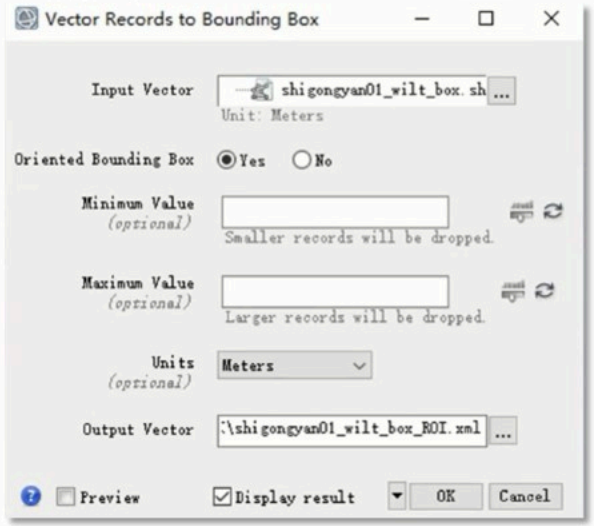


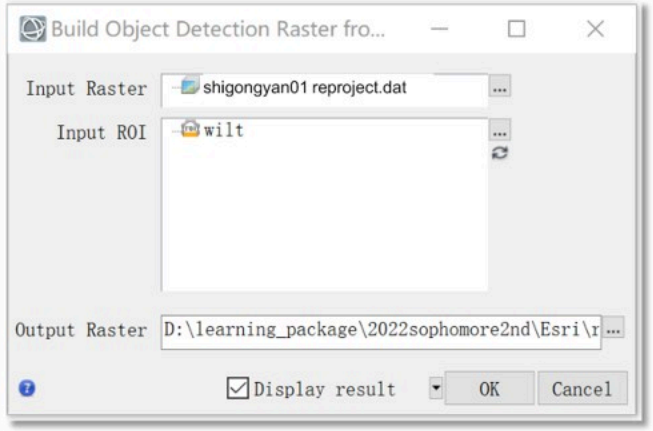
3.6.2 Deep Learning with Object Detection

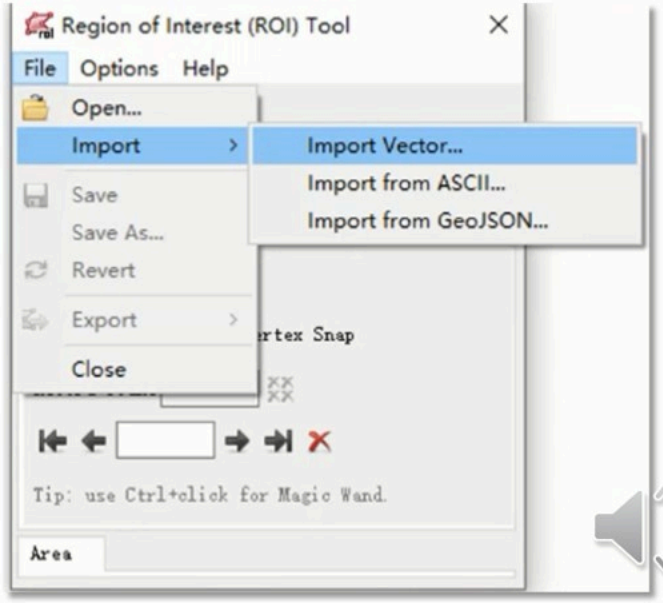
Create Label Rasters

Export the sample ROIs as vector shapefiles, convert them to rectangles, and import them as rectangular ROIs. Then create the object detection label rasters.





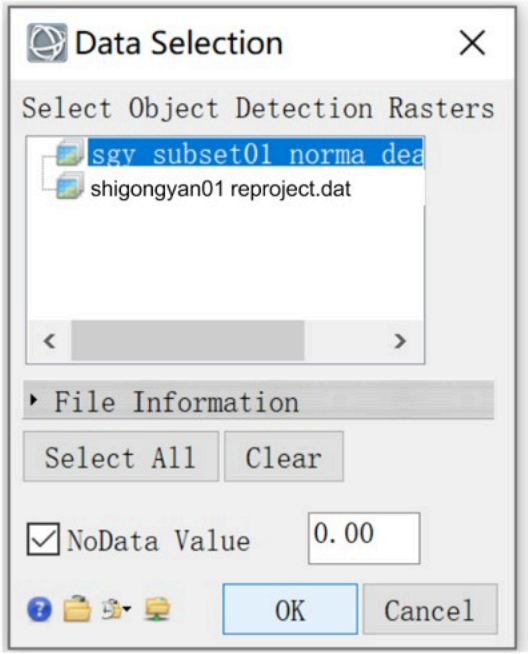




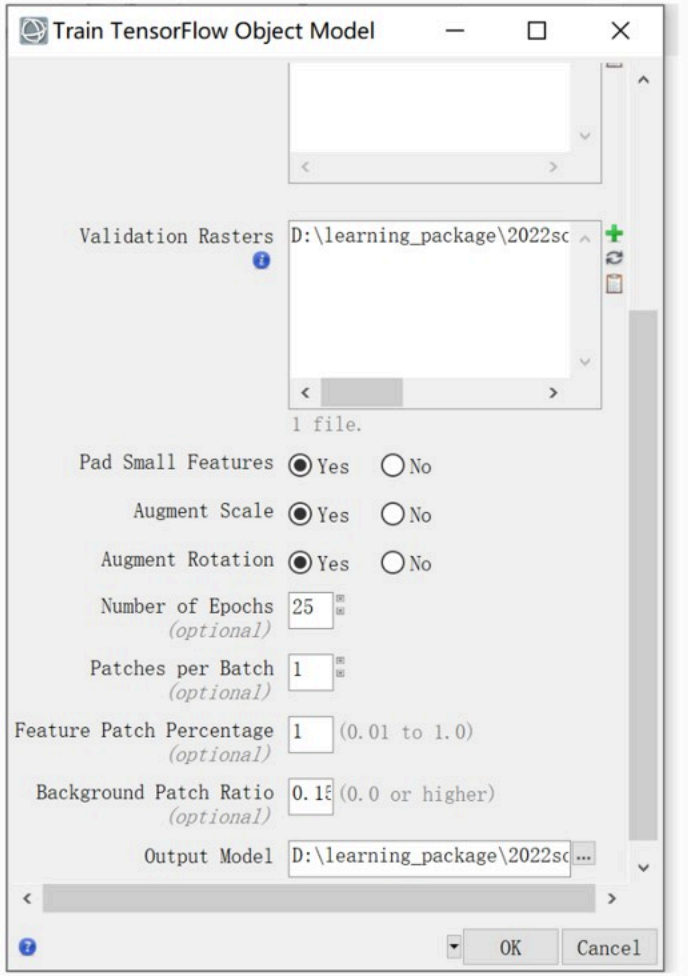
3.6.2 Deep Learning with Object Detection

Train the Deep Learning Model

Initialize a model. Input the label rasters as training rasters and also use them as validation rasters. Select NoData Value, set the number of epochs to 25, leave other parameters at their defaults, and specify the output path.



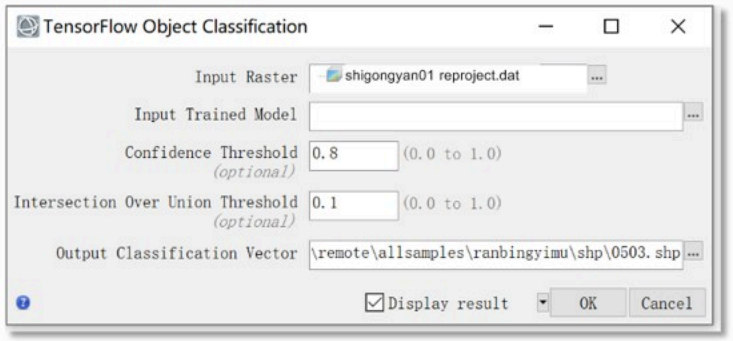


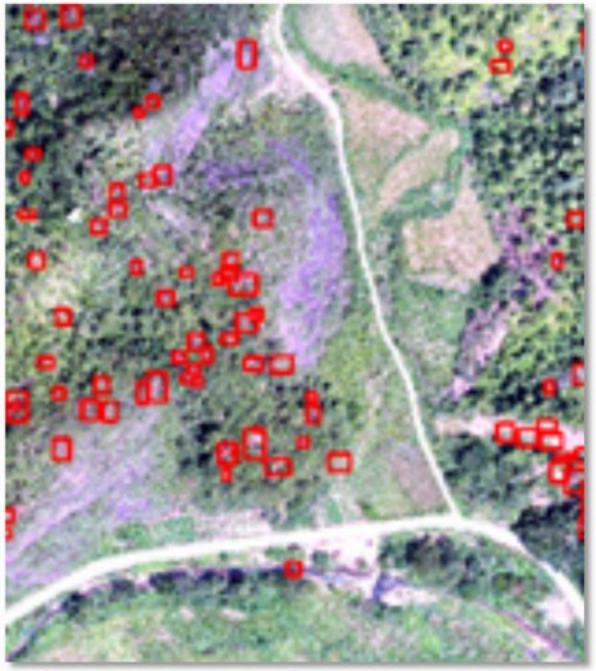


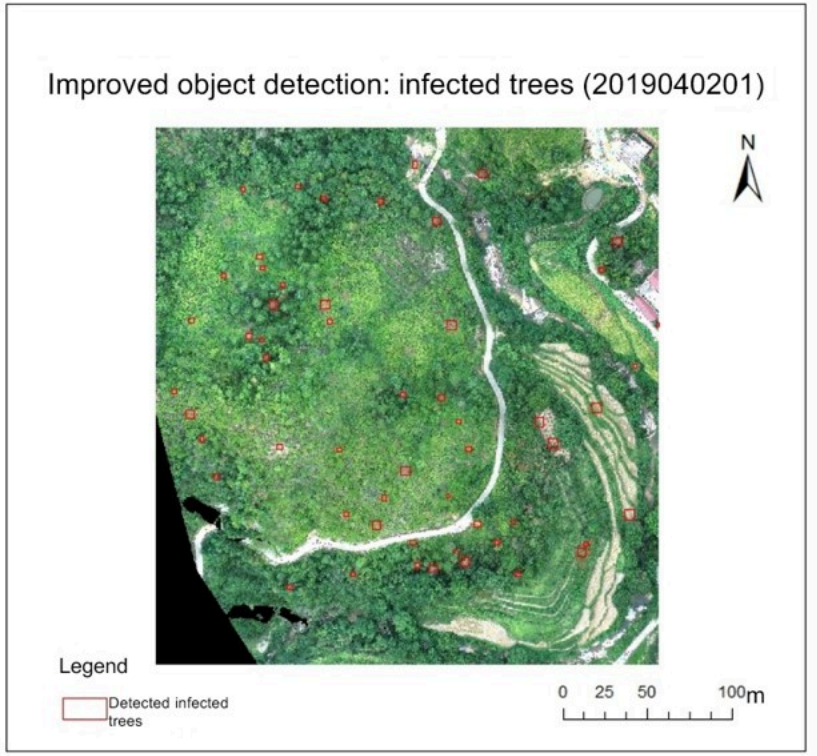
3.6.2 Deep Learning with Object Detection

Model Classification

Classify the imagery using the trained model. Input the raster to classify, set Confidence Threshold to 0.8 and Intersection over Union to 0.1, and specify the output path.





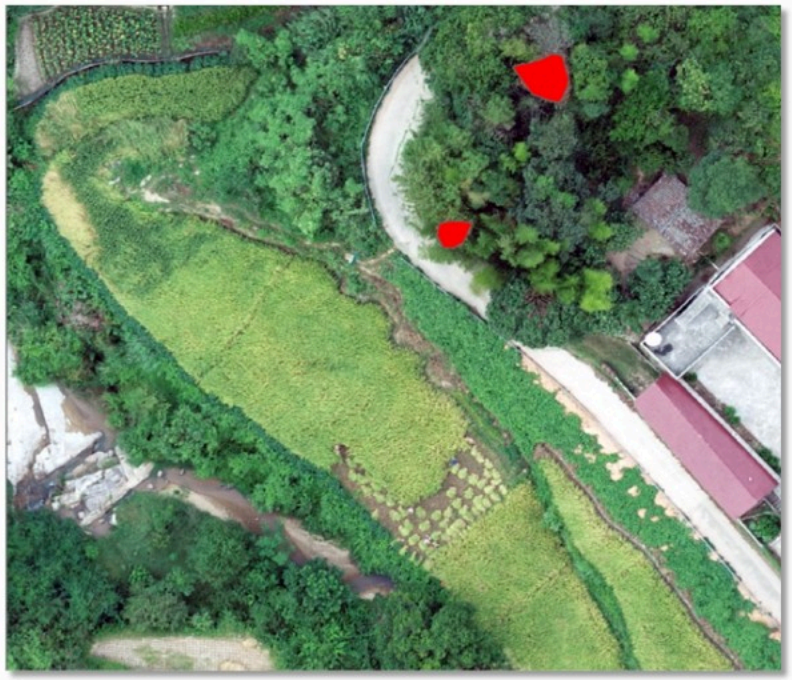


3.7 Postprocessing Model Outputs

Most false detections of infected trees came from bare ground and buildings. Extracting vegetation or pine-covered areas could remove most of these errors.



Model extraction results



Infected trees identified visually

3.7 Postprocessing Model Outputs

Multiresolution Segmentation

Multiresolution segmentation starts from individual pixels and builds larger objects from the bottom up, seeking the largest scale at which they remain similar. Similarity is judged by the objects' color and shape.

In this project, different segmentation scales produce results for different land-cover types, including the forest areas we aim to extract.

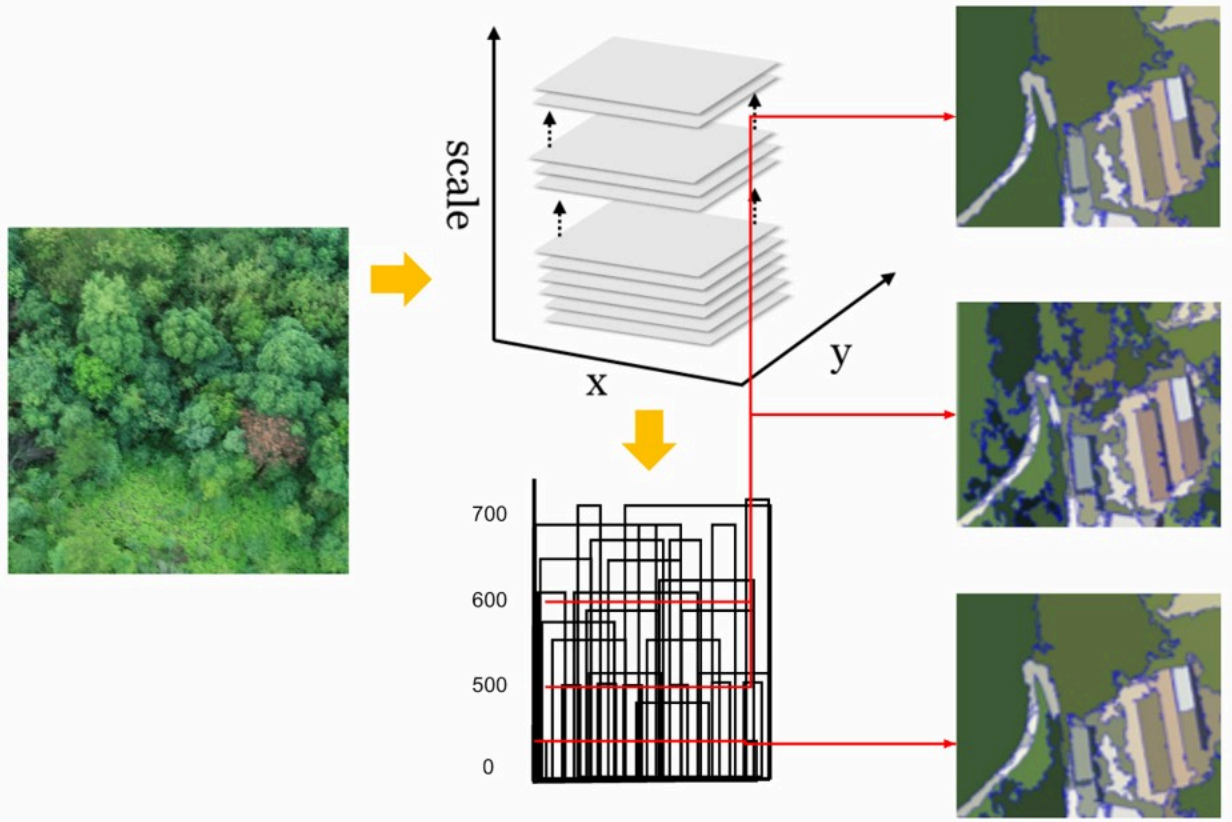
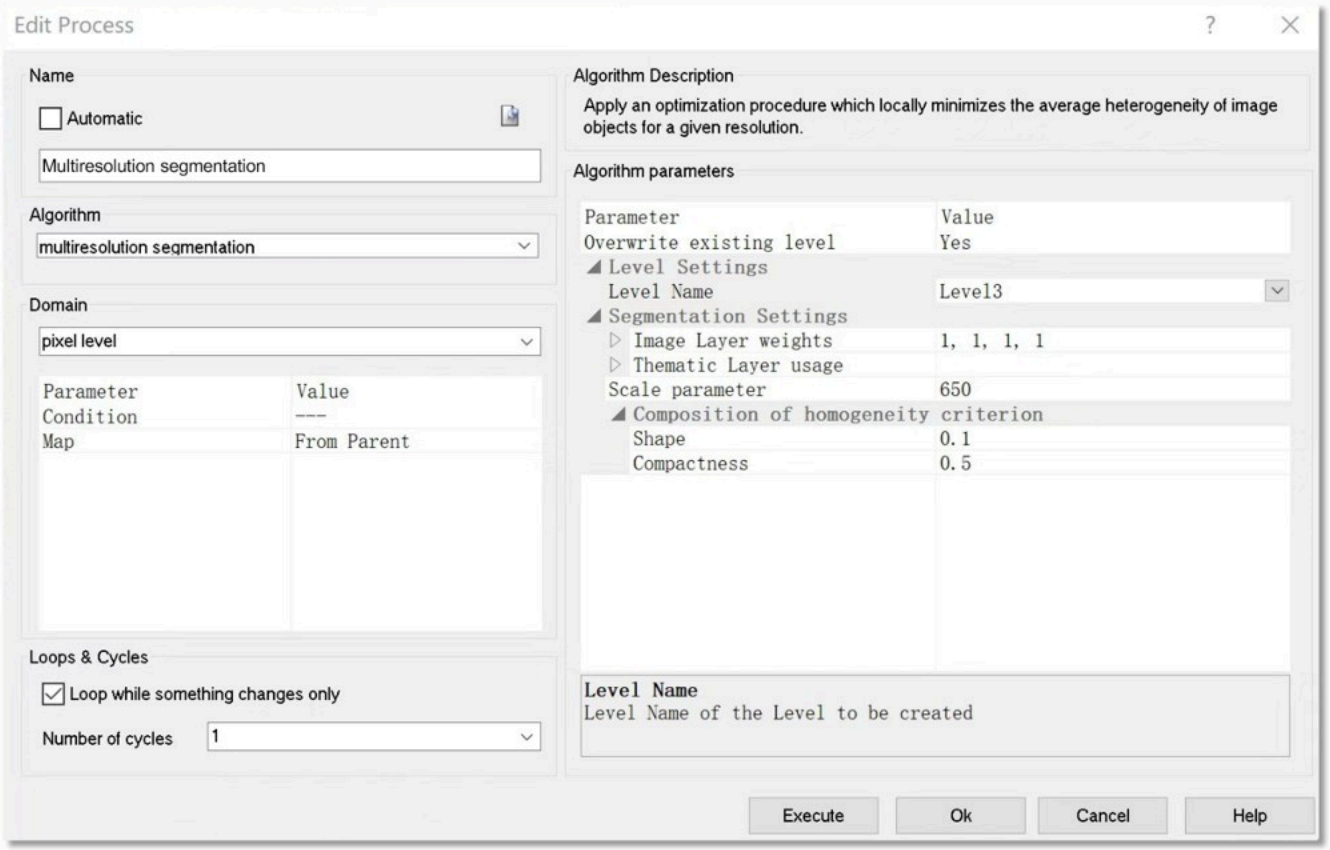
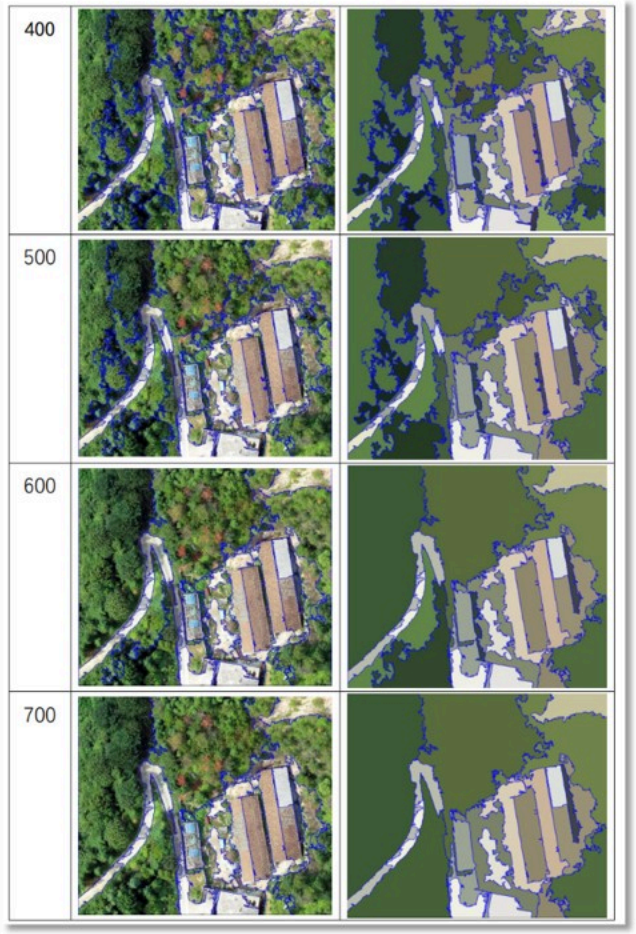


Figure: Multiresolution segmentation workflow

3.7 Postprocessing Model Outputs

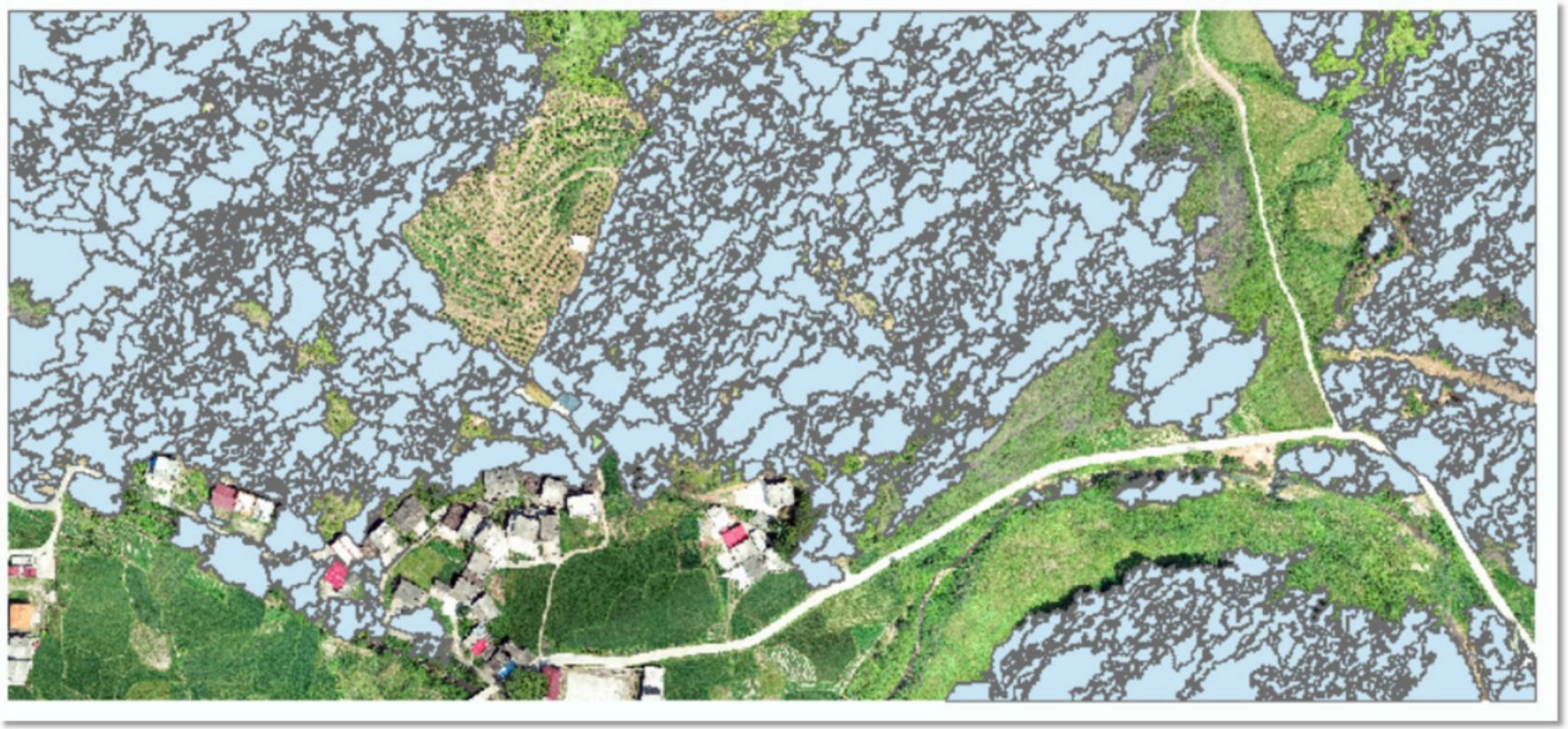
We compared different segmentation scale parameters to make the forest outlines closely match their actual boundaries. The final scale was approximately 600.





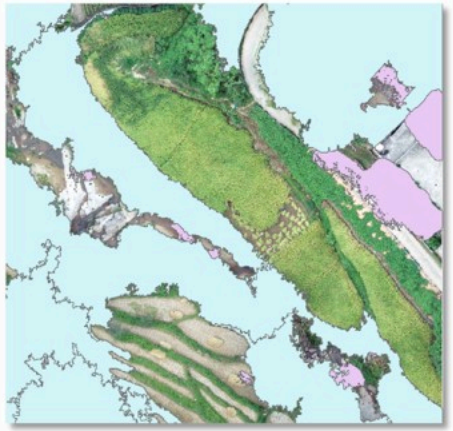
3.7 Postprocessing Model Outputs

This method extracts forest areas relatively accurately, allowing misclassified bare ground, buildings, and dry grass outside the forest to be removed. This further improves accuracy and reduces the commission error.



Forest extraction results





Misclassified land-cover features

04 Results and Analysis

4.1 Accuracy Comparison of the Two Methods

Accuracy Assessment Criteria



User's accuracy = correctly identified infected trees / all detected trees

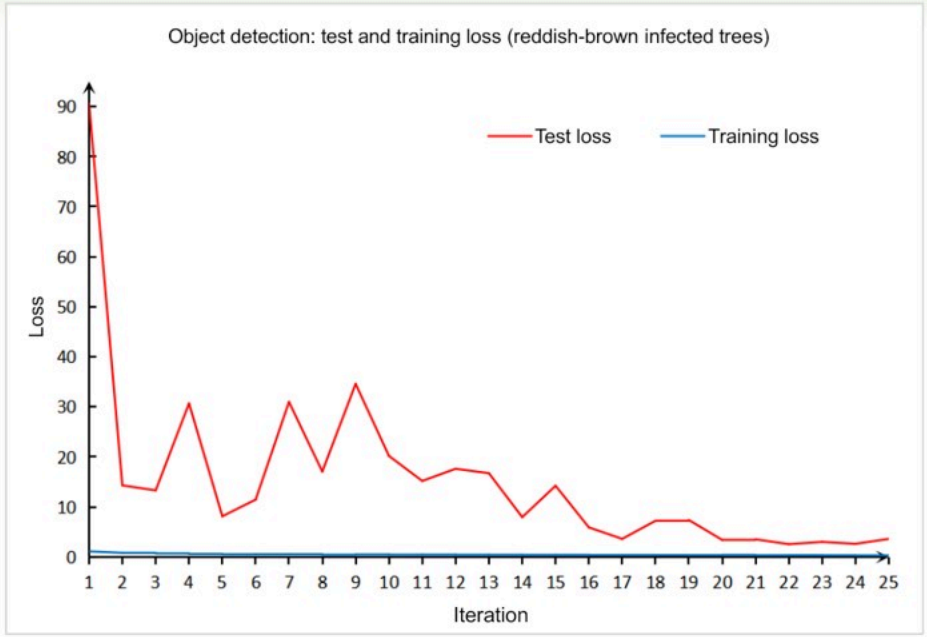
Producer's accuracy = correctly identified infected trees / actual infected trees (visual interpretation)

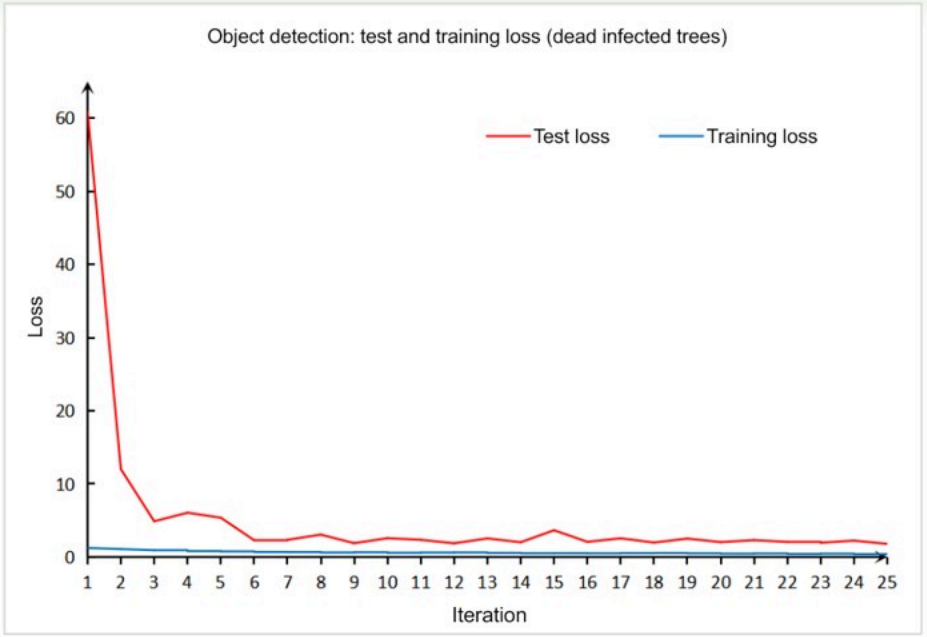
Commission error = 1 − user's accuracy

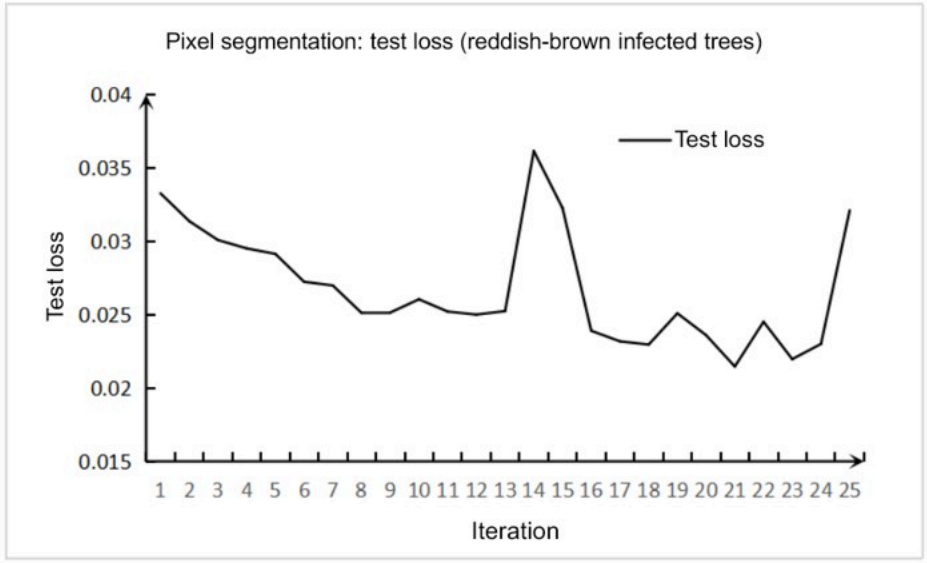
Omission error = 1 − producer's accuracy

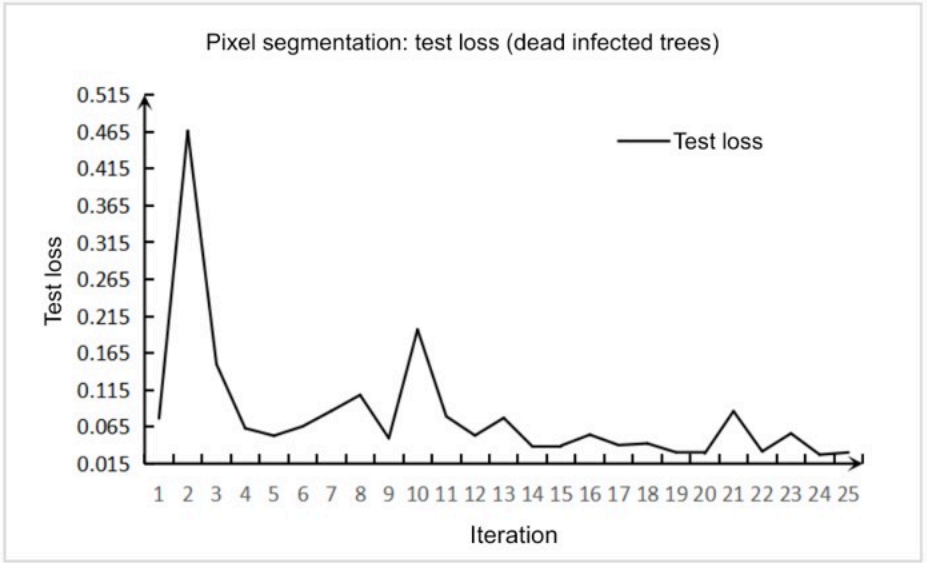
Deep learning method	User's accuracy	Producer's accuracy	Commission error	Omission error
Pixel segmentation	0.44	0.97	0.56	0.03
Object detection	0.70	0.55	0.29	0.45

4.1 Accuracy Comparison of the Two Methods



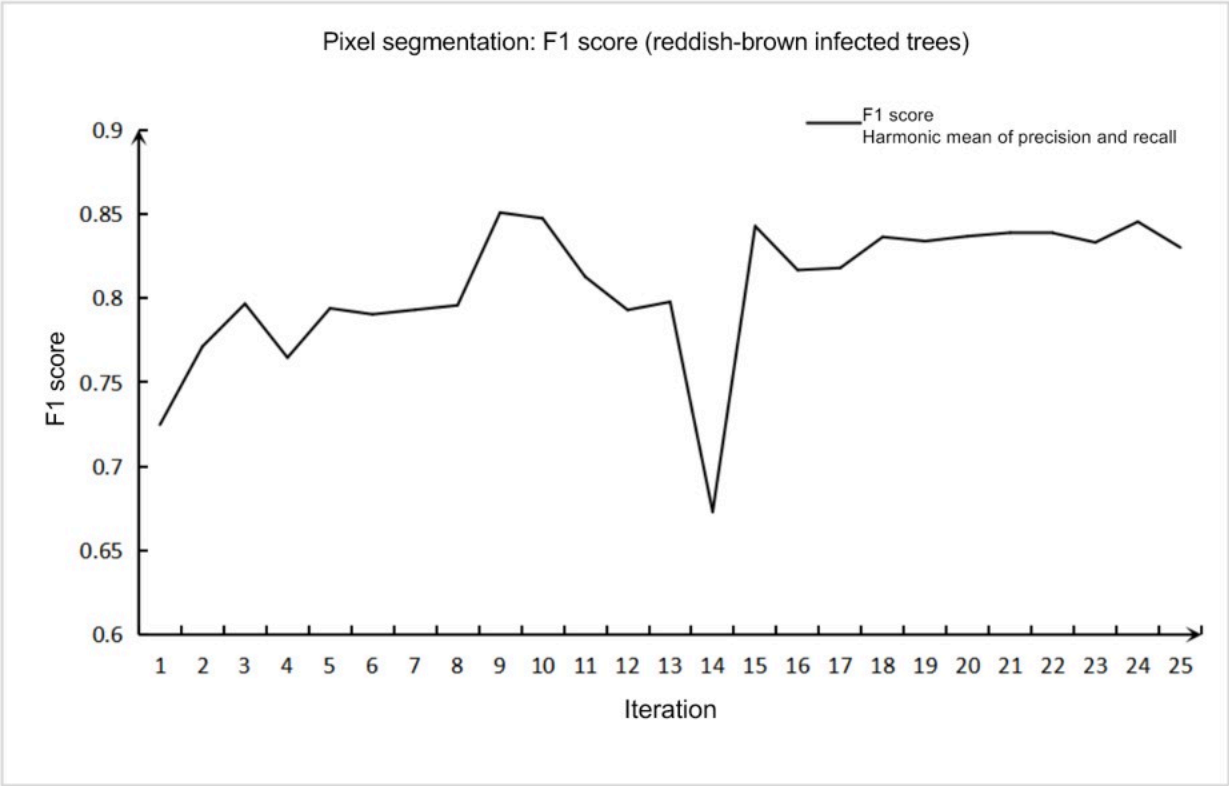


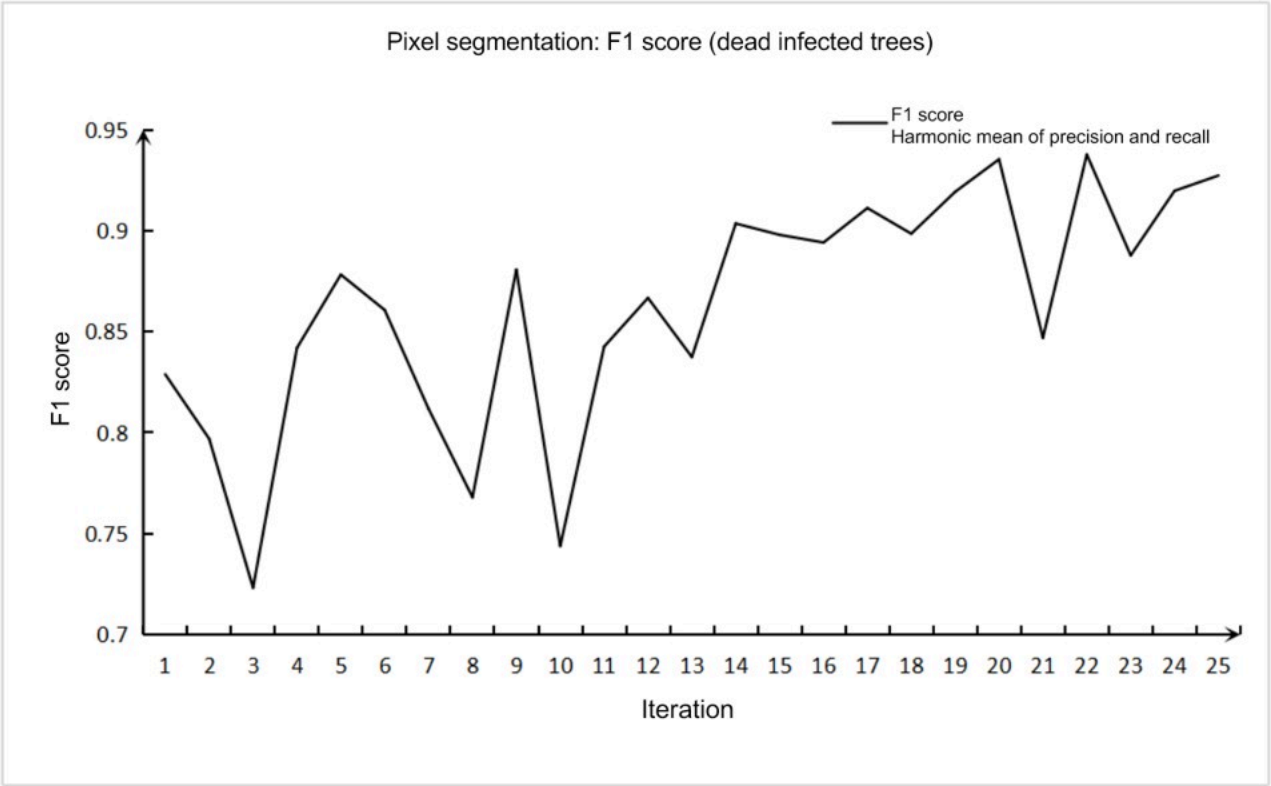




Across all four models, test loss and training loss generally decrease as the number of epochs increases.

4.1 Accuracy Comparison of the Two Methods





For both pixel segmentation models, accuracy generally increases as the number of epochs increases.

Results



This project used deep learning with pixel segmentation and object detection to extract infected and dead infected trees affected by pine wilt disease. We compared the methods through visual interpretation and accuracy assessment. Pixel segmentation proved the most practical method for extracting infected trees. It can improve the efficiency of locating pine wilt disease-infected trees and has the potential to support forestry personnel in controlling the disease.